

BARGUZIN, Russian FederationWMO: **306360**Lat: **53.617N**Lon: **109.633E**Elev: **1603**StdP: **13.86**Time Zone: **8.00 (E08)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.5	-36.1	-44.6	0.4	-39.3	-40.9	0.6	-35.9	13.1	5.2	11.6	1.6	0.2	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	86.2	64.1	82.6	63.2	79.3	62.1	67.3	78.8	65.6	76.9	63.9	75.5	4.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	93.4	71.0	61.8	87.5	69.3	59.8	81.3	68.5	32.8	79.1	31.4	76.3	30.1	75.9	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	9.9	8.3	-44.4	92.1	3.3	3.9	-46.8	94.9	-48.7	97.2	-50.6	99.4	-53.0	102.2		
(5)			-44.5	69.6	3.3	1.9	-46.9	71.0	-48.8	72.1	-50.6	73.2	-53.0	74.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	-15.4	-6.5	15.8	35.8	47.8	61.5	67.2	62.6	48.5	31.6	9.1	-10.1		
	DBStd	30.35	12.75	13.50	13.21	8.49	7.00	6.75	4.89	4.97	7.43	8.42	14.29	15.06		
	HDD50	9003	2027	1583	1059	428	127	2	0	1	115	571	1226	1864		
	HDD65	13243	2492	2003	1524	875	534	145	33	102	495	1035	1676	2329		
	CDD50	1407	0	0	0	3	60	347	534	392	71	0	0	0		
	CDD65	172	0	0	0	0	1	40	102	29	0	0	0	0		
	CDH74	2209	0	0	0	0	62	670	1122	336	19	0	0	0		
	CDH80	611	0	0	0	0	16	203	351	40	1	0	0	0		
(14) Wind	WSAvg	3.1	2.1	3.0	3.5	4.2	4.5	3.6	3.2	3.0	3.3	2.8	2.3	1.8		
(15) Precipitation	PrecAvg	13.1	0.5	0.2	0.2	0.4	0.6	1.4	2.9	2.3	1.5	0.8	1.3	1.2		
	PrecMax	20.2	1.7	0.7	1.1	1.1	1.9	4.3	8.1	5.2	4.3	2.1	3.4	2.9		
	PrecMin	7.4	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.6	0.2	0.0	0.4	0.0		
	PrecStd	3.0	0.4	0.2	0.2	0.3	0.4	1.1	1.9	1.3	1.0	0.5	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	32.1	48.2	68.5	81.2	90.3	91.2	83.6	77.3	57.5	43.0	34.0		
		MCWB	18.6	26.8	36.9	48.8	57.1	61.5	65.3	63.6	58.9	45.8	36.1	30.5		
	2%	DB	15.2	25.6	43.3	60.6	73.4	85.5	87.6	80.7	70.6	51.4	38.2	25.2		
		MCWB	13.2	22.7	34.1	44.7	53.2	62.2	65.6	62.7	55.6	42.2	33.7	22.7		
	5%	DB	9.7	18.8	38.8	55.2	68.4	81.6	84.1	77.9	66.2	47.8	34.4	20.8		
		MCWB	8.0	16.3	31.5	41.6	50.3	60.9	64.8	62.1	53.9	39.9	29.9	19.0		
	10%	DB	4.4	14.0	34.7	50.5	64.2	77.5	80.6	74.8	61.6	44.5	29.8	13.4		
		MCWB	3.1	11.7	28.7	39.0	48.1	60.2	63.9	61.1	51.4	38.2	26.5	11.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	26.7	38.2	48.5	58.0	67.3	70.2	67.5	59.4	47.1	37.3	31.1		
		MCDB	21.5	31.7	47.4	66.9	79.4	81.0	82.6	77.9	71.1	55.5	41.3	33.6		
	2%	WB	13.2	22.5	34.5	45.3	54.3	64.3	68.2	65.5	57.4	43.7	34.2	23.0		
		MCDB	15.2	26.2	42.9	59.3	71.2	79.2	80.2	75.4	67.8	49.6	38.1	25.3		
	5%	WB	7.9	16.2	31.4	42.2	51.4	62.9	66.7	63.9	55.1	40.9	30.0	18.8		
		MCDB	9.5	18.3	38.3	53.9	65.6	77.6	78.2	74.0	64.1	47.0	34.1	20.7		
	10%	WB	3.0	11.8	28.7	39.5	48.7	61.3	65.3	62.2	52.7	37.9	26.5	11.9		
		MCDB	4.3	13.9	34.5	49.0	62.8	74.1	76.7	72.1	60.8	43.9	30.0	13.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	19.2	21.6	20.4	23.2	24.2	22.3	20.6	18.9	16.9	15.1	14.0		
		MCDBR	22.8	18.6	22.7	29.7	34.2	32.8	30.6	27.4	27.3	22.7	16.8	17.4		
		MCWBR	21.0	15.5	14.5	15.7	16.8	12.9	12.3	12.0	13.3	13.6	13.1	15.8		
	5% WB	MCDBR	22.9	18.9	22.2	28.1	29.9	27.4	24.3	23.4	23.8	20.9	16.9	17.4		
		MCWBR	21.2	16.0	14.5	15.3	15.1	12.0	11.0	11.5	11.9	13.3	13.9	15.9		
(40) Clear-Sky Solar Irradiance	taub		0.272	0.306	0.335	0.388	0.453	0.476	0.446	0.406	0.355	0.320	0.291	0.270		
	taud		2.041	2.002	2.011	2.057	2.009	2.063	2.219	2.306	2.392	2.335	2.097	2.002		
	Ebn at Noon		210	241	262	264	252	247	253	258	259	241	200	181		
	Edh at Noon		23	34	42	46	51	49	42	36	29	24	22	20		
(44) All-Sky Solar Radiation	RadAvg		309	620	1070	1493	1798	1931	1760	1472	1059	609	299	205		
	RadStd		26	37	44	94	112	118	151	69	84	60	30	17		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	-3.28	-4.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page